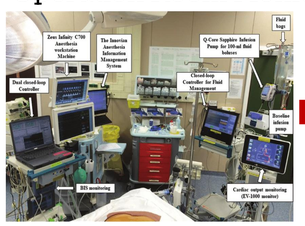
Beyond Euclidean Summaries: Online Change Point Detection for Distribution-Valued Data

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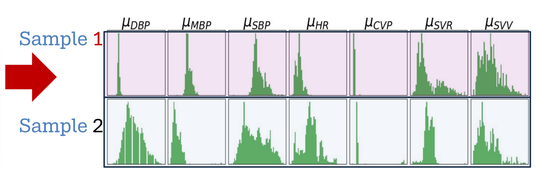
1. Motivation I: The distribution-valued sample

Modern sensing and observational systems produce batches of observations at each time step. In these systems, each observation at time t is a **distribution-valued sample**.

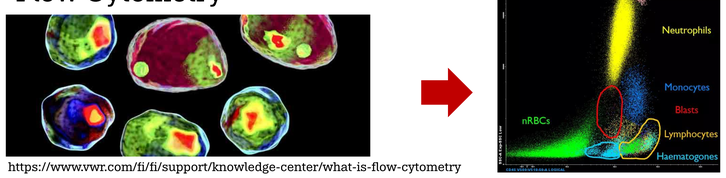
Operation Room Monitoring



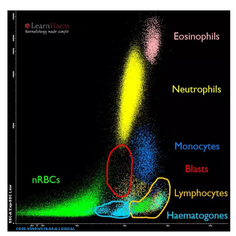
Distribution-value Samples



Flow Cytometry



https://www.vwr.com/fi/fi/support/knowledge-center/what-is-flow-cytometry

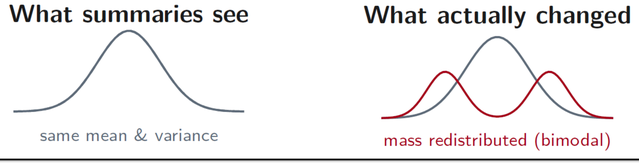


One Sample (thousands of cells) of One Patient

2. Motivation II: Literature ignore geometry

Our Objective: detect an abrupt change in a **distribution-valued data stream** as quickly as possible while controlling false alarms.

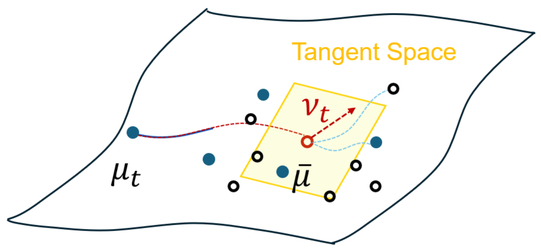
Literature	Limitation
SPC / Hotelling (T^2)	Low-order moment cannot capture distributional change
Kernel / MMD / NEWMA	Scalar discrepancy score
Fréchet / object-valued CPD	Metric summaries can ignore direction of distributional shift
Wasserstein CPD	Often limited interpretability / scope (i.e., univariate setting)



3. Our core contribution: an intrinsic distribution-valued CPD (IDD) scheme

- We formulate online Change Point Detection (CPD) for a distribution-valued process $\{\mu_t\}_{t\geq 1}$ and detect changes in its law by transporting each μ_t to pre-change Fréchet barycenter $\bar{\mu}$.

→ Directly working with Wasserstein geometry



Monitoring Distribution-valued Samples μ_t

↓

Monitoring Optimal Transport (OT) Maps $T_{\bar{\mu}}^t = T_{\bar{\mu},\mu_t\#}\bar{\mu}$

↓

Monitoring OT displacement field: $v_t := T_{\bar{\mu}}^t - \mathrm{Id}$

- We develop a two-chart online monitoring procedure leveraging Multivariate Functional PCA (MFPCA) to derive Hotelling's T2 and Squared Prediction Error (SPE) statistics in the Wasserstein tangent space.

Hotelling's T^2 (in-subspace)

$$T_{t,K}^2 = \sum_{m=1}^{K} \frac{\hat{\xi}_{tm}^2}{\hat{\lambda}_m}$$

Detects shifts *along* the dominant pre-change modes of variation.

SPE / Q (out-of-subspace)

$$\mathrm{SPE}_t = \left\|(I - \hat{P}_K)\Delta_t\right\|_H^2$$

Detects *new* geometry the reference period never showed.

- Theoretical guarantees
  - ε-Isometry guarantee

Theorem 3.14 (ε-Isometry)

Let $\mathcal{X} \subset \mathbb{R}^d$ and the covariance kernel $\mathcal{K}$ satisfy Prop. 3.13 (Lipschitz kernel). Let $\{\lambda_m\}_{m\geq 1}$ be the eigenvalues of the covariance operator Γ in non-increasing order. Then there is a constant $A_{\mathcal{X}}$ depending only on d with

$$\sum_{m>K} \lambda_m \leq A_{\mathcal{X}}\, C_{\mathcal{K}}\, K^{-1/d}$$

Consequently, to achieve an ε-isometry in mean square (relative reconstruction error $\leq \varepsilon^2$), it suffices that

$$K \geq \left(\frac{A_{\mathcal{X}}\, C_{\mathcal{K}}}{\varepsilon^2\, \mathrm{tr}(\Gamma)}\right)^d$$

→ This ε-isometry enables geometry-preserving finite-dimensional embeddings for scalable CPD.

- Asymptotic behavior of the monitoring statistics

Proposition 3.9 (Asymptotic null distribution)

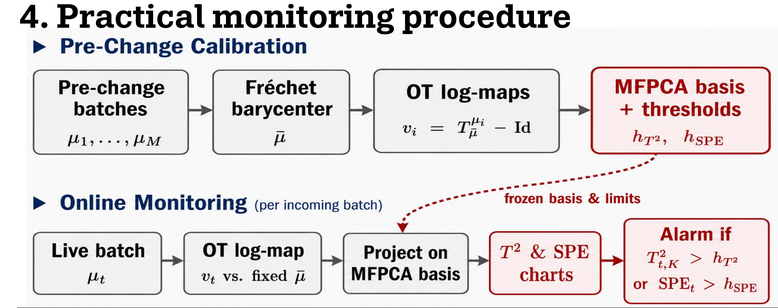
As the pre-change size $n_0 \to \infty$, for fixed K,

$$T_{t,K}^2 \xrightarrow{d} \chi_K^2, \qquad \mathrm{SPE}_t \xrightarrow{d} \sum_{m>K} \lambda_m Z_m^2, \qquad Z_m \sim \mathcal{N}(0,1).$$

Empirical quantiles of the calibration set give data-driven thresholds.

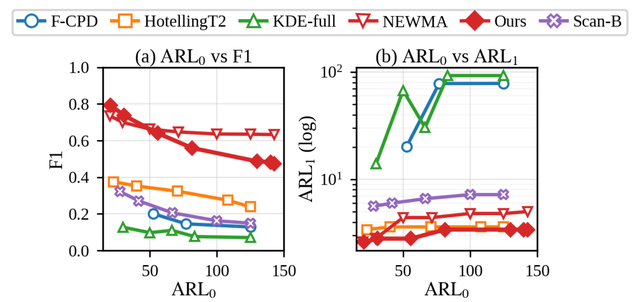
→ This asymptotic behavior enables statistically principled thresholds for T2 and SPE charts, allowing scalable online CPD with controlled false alarms.

4. Practical monitoring procedure



5. Real world case studies

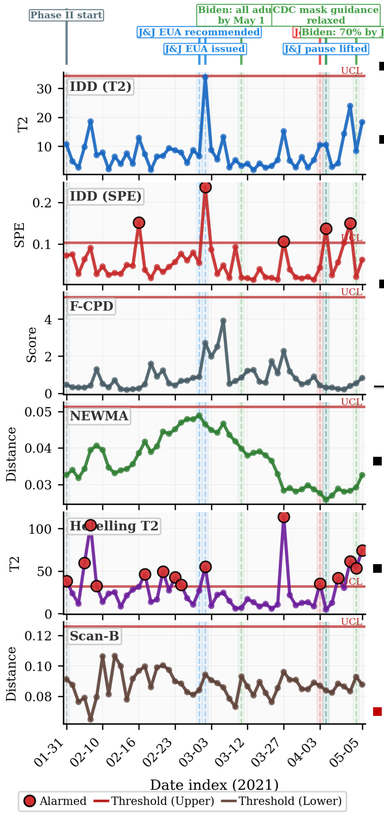
- Abnormal Cell Detection (FlowCAP-II AML detection)



Flow-cytometry samples are distributions in $P_2(R^7)$.

IDD: F1 ≈ 0.75, ARL_1 ≈ 1, precision ≈ 0.99.

- Public Opinion Monitoring on COVID Vaccine



- IDD(SPE): 5 targeted alarms.
- Aligned with J&J EUA and pause-lifting discourse; May 3 precedes Biden July-4 vaccination goal.
- Baselines: zero alarms or many unaligned alarms.

6. Takeaways

- Distribution-valued streams need geometry-aware detection.
- A distributional-free online CPD is proposed to monitor distribution-valued samples in the 2-Wasserstein space.
- Future Work: Expediate the estimation of barycenter and optimal transport plans.